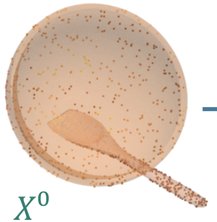
(a) Structured Interaction Dynamics

Initial metric scene



Reference I^0



X^0

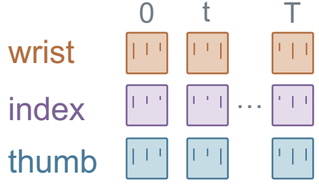
Point encoder

Hand trajectory



$H^{0:T}$

Joint-aware action encoding



Joint encoder

joint + hand + time

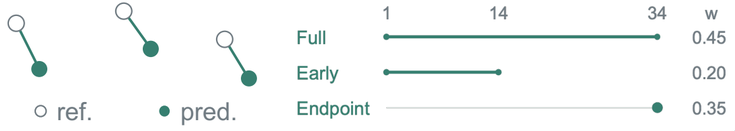
Object Transformer

Metric-aligned refinement

Valid moved points

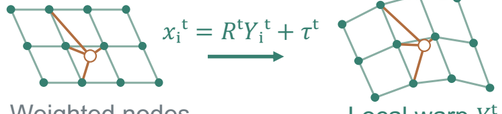
Metric distance (mm)

$\|\Delta x_{ref}\| > \delta$ / frame



Global pose R^t, τ^t

Object-local graph



Metric 3D forecast



Rigid



Deformable

Persistent points + confidence Q

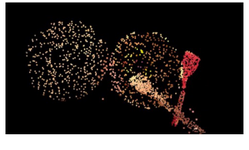
(b) Confidence-gated Visual Readout

Reference I^0

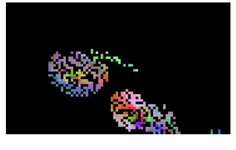


Project & transport

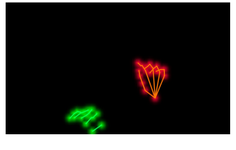
Projected geometry + Q



Appearance transport



Hand conditioning



VACE
Conditioning adapters

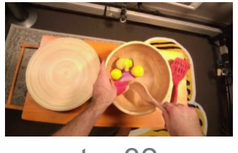
Video diffusion
Wan

Future RGB



t = 16

…



t = 32